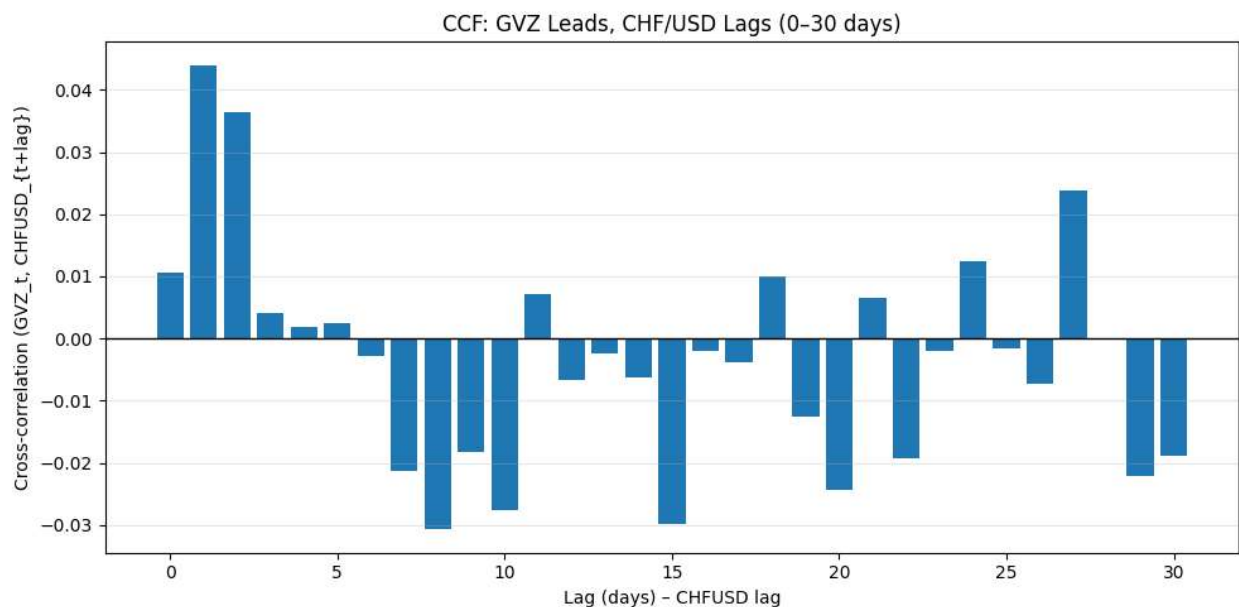


Thesis: To US investors both Gold and Swiss Francs are used as haven investments. But what happens when one of the two becomes unstable? My thesis is that when gold volatility spikes, US investors looking to invest in safer assets will buy more Francs. In essence, once GVZ (gold volatility index) spikes, CHF/USD will increase after a certain lag.

Preliminary EDA:

Firs I ran a CCF test on GVZ returns to CHF/USD returns at lags 0-30

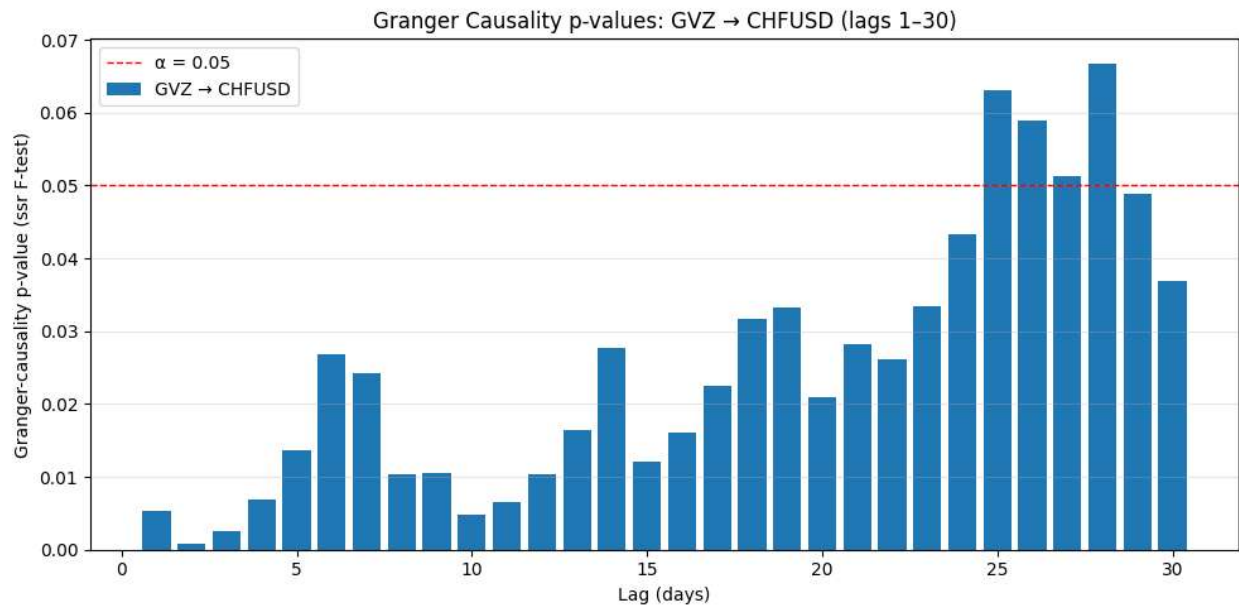


Highest correlation: 0.0440 at lag 1 days

Lowest correlation: -0.0307 at lag 8 days\

The short lags align with my thesis as gold vol information is public and therefore should be quickly priced into something liquid like FX. Though the correlation is low, that is somewhat expected as gold vol should only slightly predict the FX rate.

This however does not contain statistical evidence backing my thesis so I next did a granger causality test for lag 1-30.

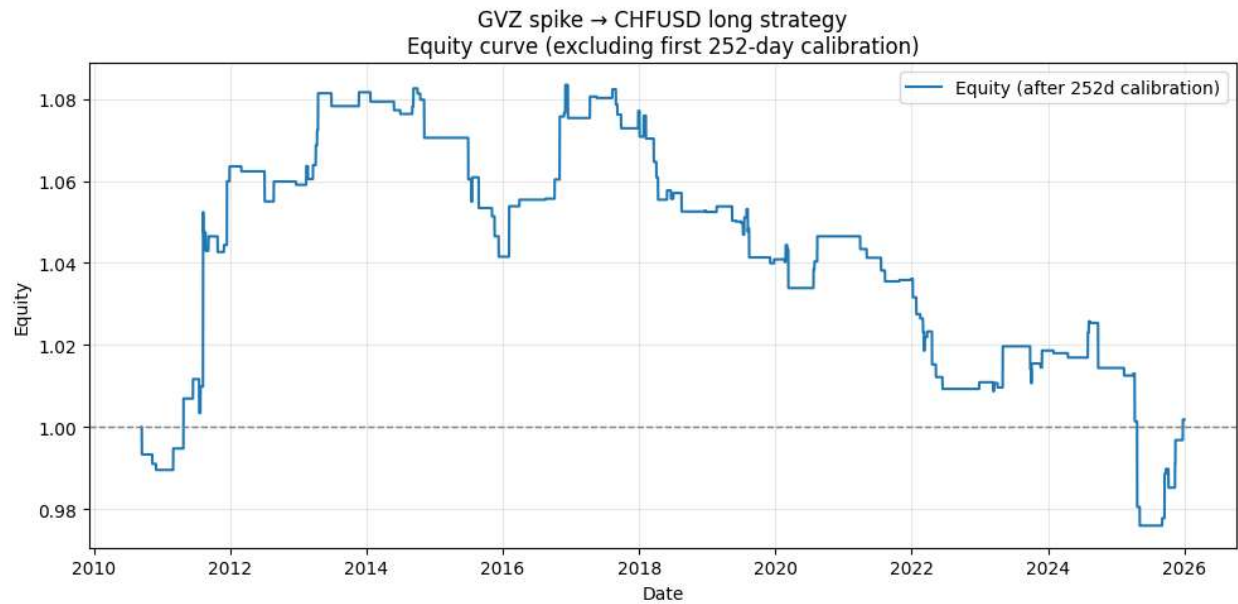


Minimum p-value: 0.000842 at lag 2 days

The p-value is extremely low at a lag of two but also significant over low and medium term lags.

Preliminary Backtest:

With this I ran a backtest. For the first backtest I used a lag of two at looked at when returns of GVZ were significantly high and longed (z-score greater than 2 based off the last year rolling) and exited once the z-score went under this threshold



Cumulative return (after 252d calibration): 0.18%

Annualized return: 0.02%

Annualized volatility: 1.56%

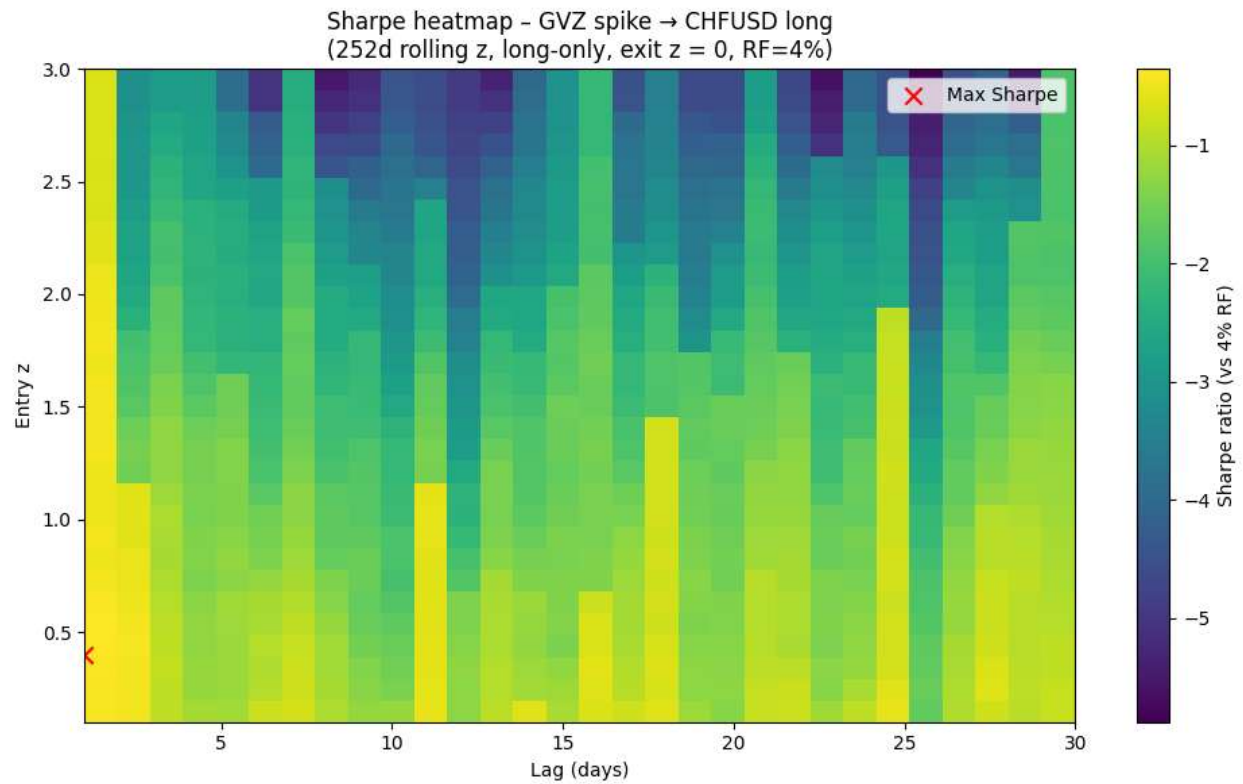
Total trades: 202

Days in trade: 211

Sharpe ratio vs 4% RFR: -2.507

The returns weren't very good, but I thought it might be due to filtering Gold vol too much with a 2 z-score as seen by the very infrequent trading.

To fix this I did a Sharpe Heatmap to test for optimal lag as well as optimal z-score.



The best Sharpe came at a lag of 1 day with a Sharpe of -0.353

I ran a backtest with these parameters



Cumulative return (after 252d calibration): 47.08%

Annualized return: 1.92%

Annualized volatility: 5.74%

Total trades: 902

Days in trade: 1159

Sharpe ratio vs 4% RFR: -0.353



Parameters: entry $z = 0.4$, exit $z = 0.4$, lag = 1 days

Cumulative return (test period): 9.32%

Annualized return (test period): 1.32%

Annualized volatility (test period): 4.02%

Total trades (test period): 277

Days in trade (test period): 371

Sharpe ratio vs 4% RFR (test period): -0.649

The Sharpe for both is negative, but trades less than half the time for each. Also in the testing data there is one day that spikes, which seems to be correlated with the Swiss Bank taking out the exchange rate floor with the EUR

Next Steps:

Given that in 2015, there was a huge single day jump for the reason above, my first next step would be to look at data and test on data after that date. This would force me to change the date splitting my test and training data (probably to 2022).

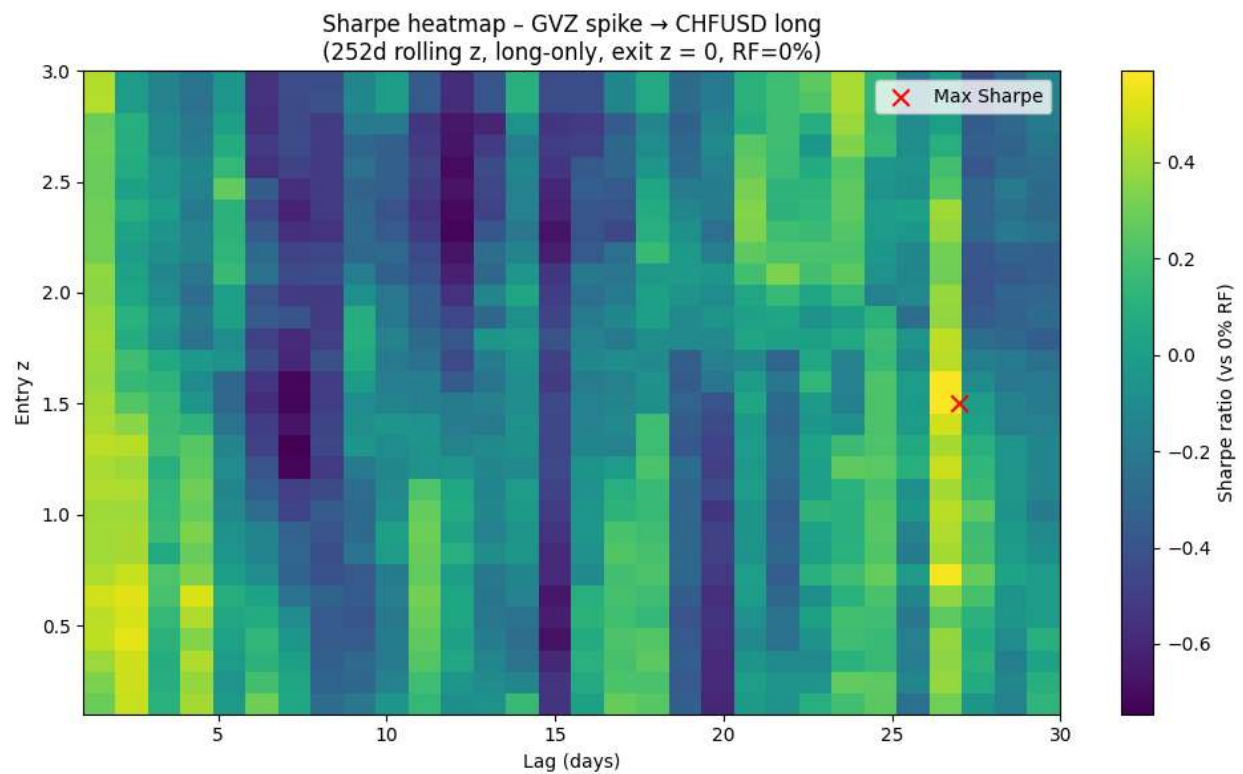
Also there may be occurrences that simultaneously push both gold vol and CHF/USD up such as high VIX. I would like to filter out these occurrences by filtering out data based on daily VIX levels/VIX returns

Finally, I believe that the 252 day rolling window might not be as great for filtering out signal compared to a smaller rolling window. I would like to test other rolling window.

Besides the above, I think more tests are needed to verify the relationship between the two

Update:

Removing RFR from the Sharpe Map gives the following heatmap.

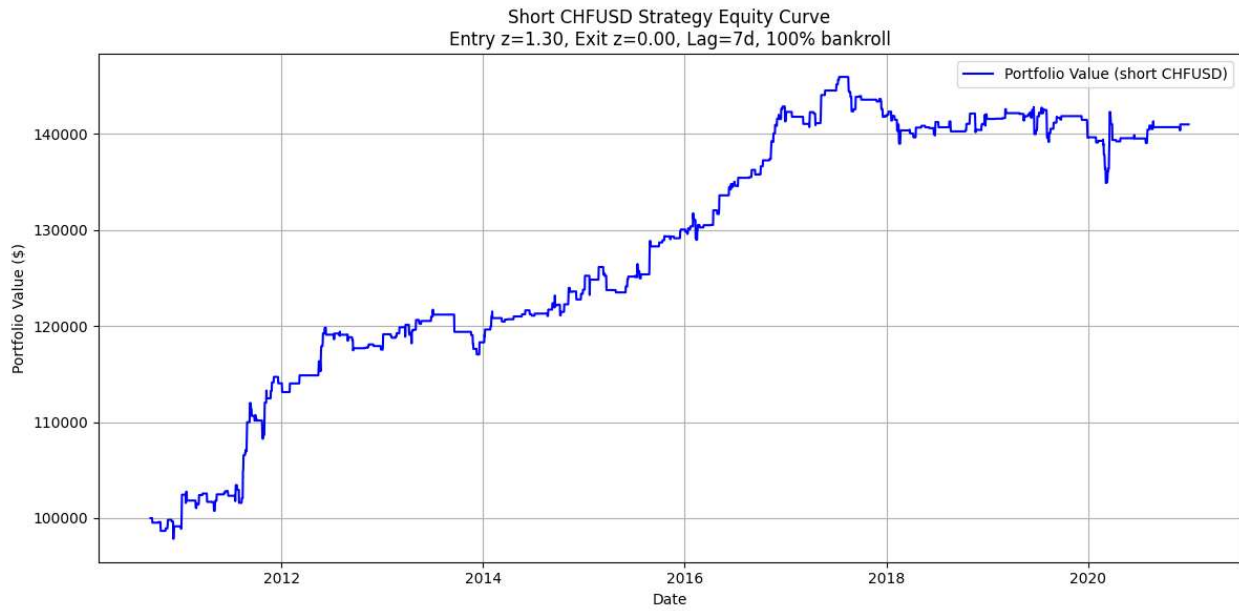


Best Sharpe (vs 0% RF): 0.590 at entry $z = 1.50$, lag = 27 days

Worst Sharpe (vs 0% RF): -0.746 at entry $z = 1.30$, lag = 7 days

Reversion the backtest on the strategy yields the following returns

Training set:



Entry z (GVZ): 1.30

Exit z (GVZ): 0.00

Lag: 7 days

Total trades (days in position): 467

Days in position: 467

Cumulative return: 40.99%

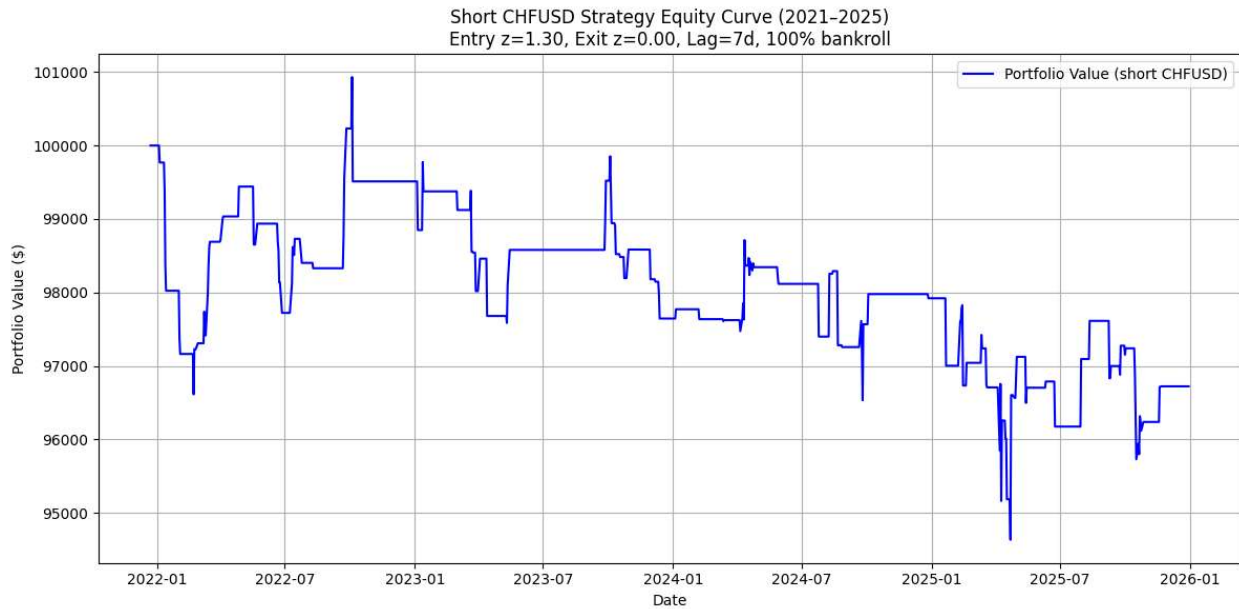
Annualized return: 2.33%

Annualized volatility: 3.15%

Sharpe vs 0% RFR: 0.74

Sharpe vs 4% RFR: -0.53

Testing Set:



Entry z (GVZ): 1.30, Exit z : 0.00, Lag: 7 days

Total trades (days in position): 138

Days in position: 138

Cumulative return: -3.28%

Annualized return: -0.80%

Annualized volatility: 3.10%

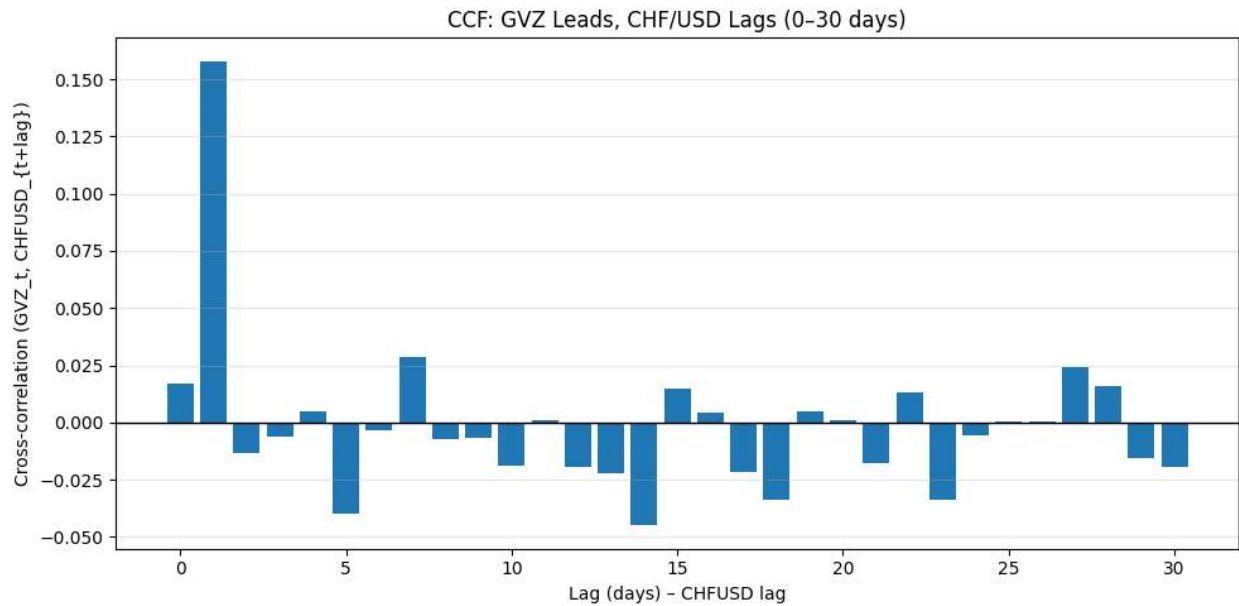
Sharpe vs 0% RFR: -0.26

Sharpe vs 4% RFR: -1.55

2016-2021 training

Due to the price floor existing during my previous training period, I rearranged my training and testing data to have training start in 2016 and end at 2021.

I first ran a CCF

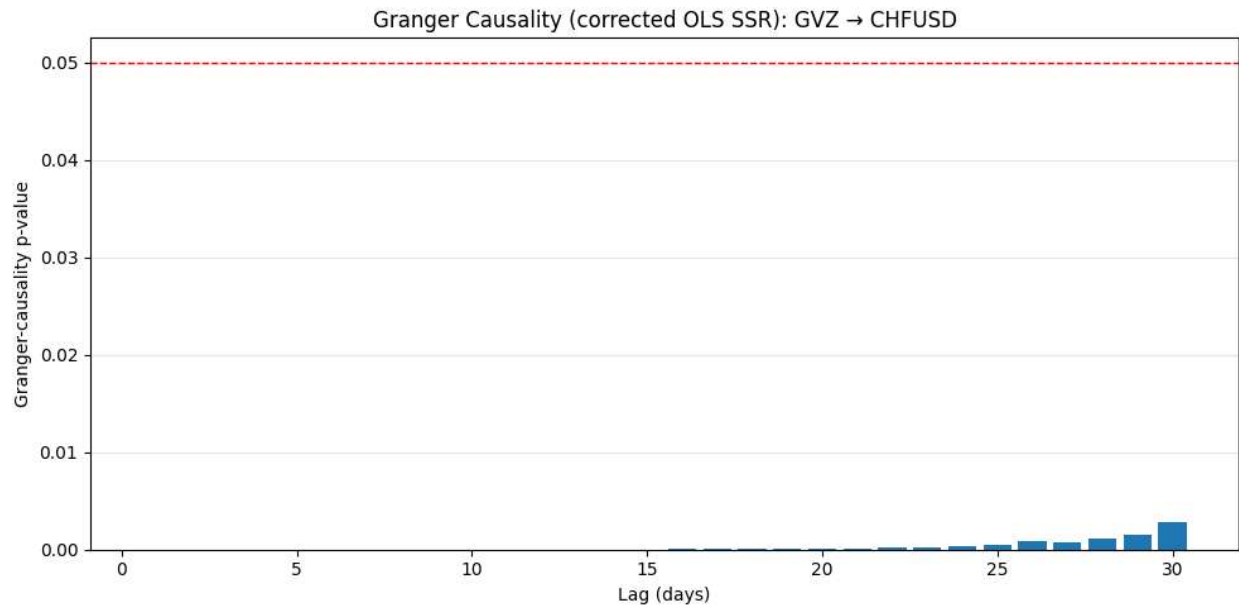


Highest correlation: 0.1580 at lag 1 days

Lowest correlation: -0.0450 at lag 14 days

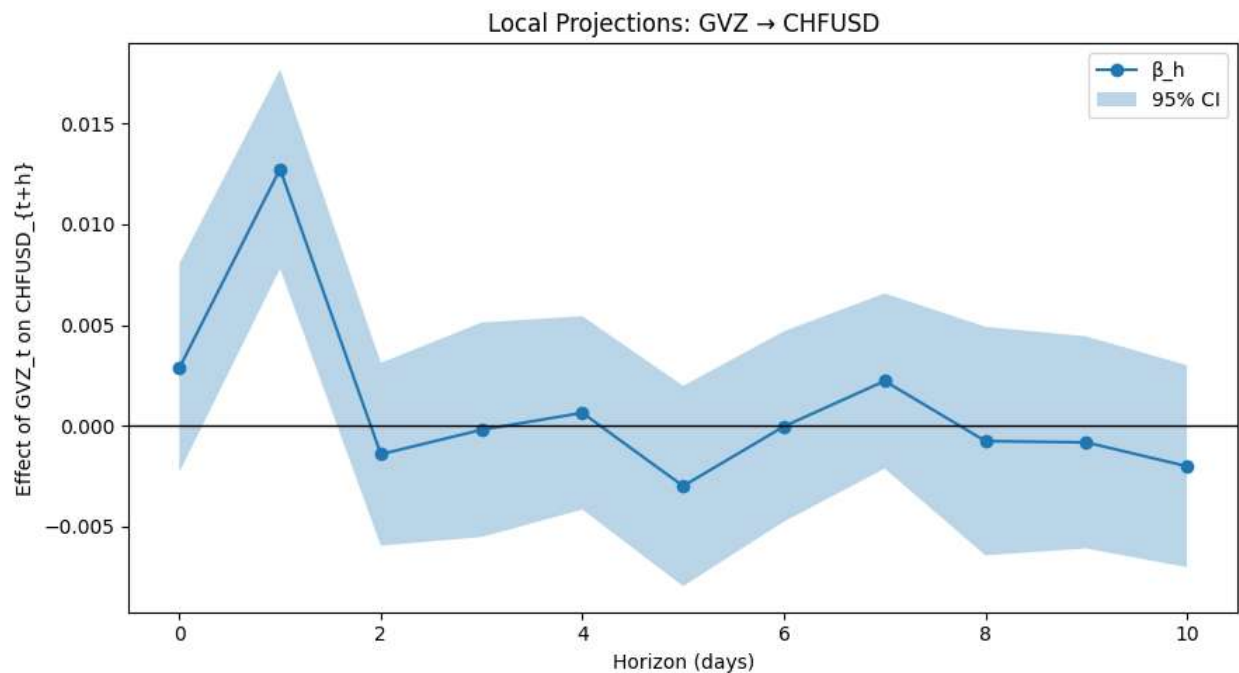
For the same reason as before the short time frame aligns with my thesis

I then ran a Granger Causality test



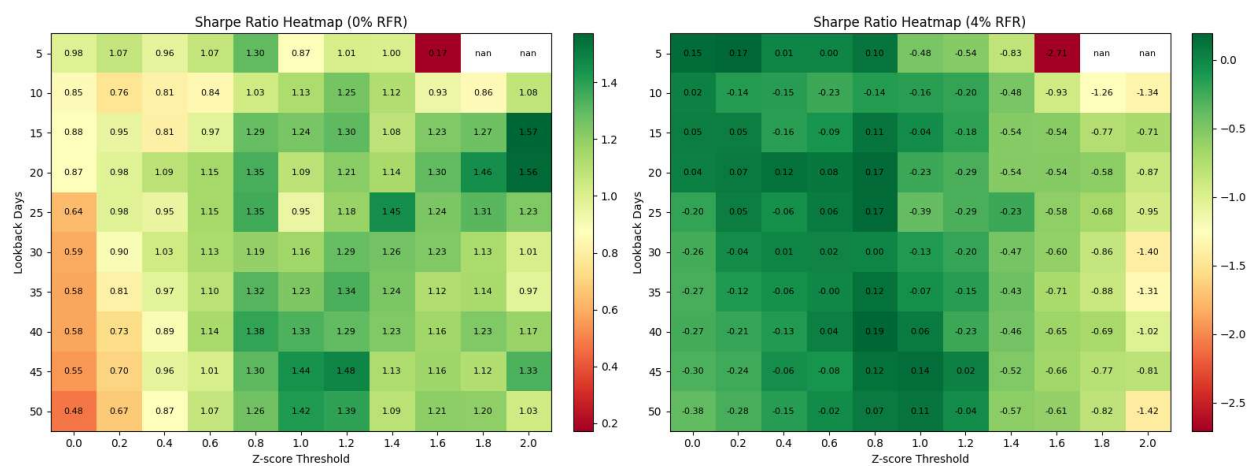
Minimum p-value: 0.000000 at lag 1

I believe that there might be cases where a third variable may be simultaneously affecting both and also do to changes in regimes. For this reason I believe these results of the Granger test may be flawed and decided to switch to using a linear projection test. I tested this with a lag of 5.



Horizon 1: Beta = 0.012743, stand error= 0.002523, t-score= 5.050322, p-value = 4.410670e-07

These results demonstrate that GVZ can be used to forecast CHFUSD returns with a horizon of 1 day. Based off this I ran a sharpe heatmap to optimize z-score of GVZ moves to enter a position as well as optimized for lookback period to address the issue of lookback period. I graphed against both 0% and 4% rfr



BEST 0% RFR:

BEST 0% RFR:

Lookback: 15 days, Z-threshold: 2.00, Sharpe: 1.57, Trades: 56.0, Days in position: 56,

Cumulative Return: 18.34%

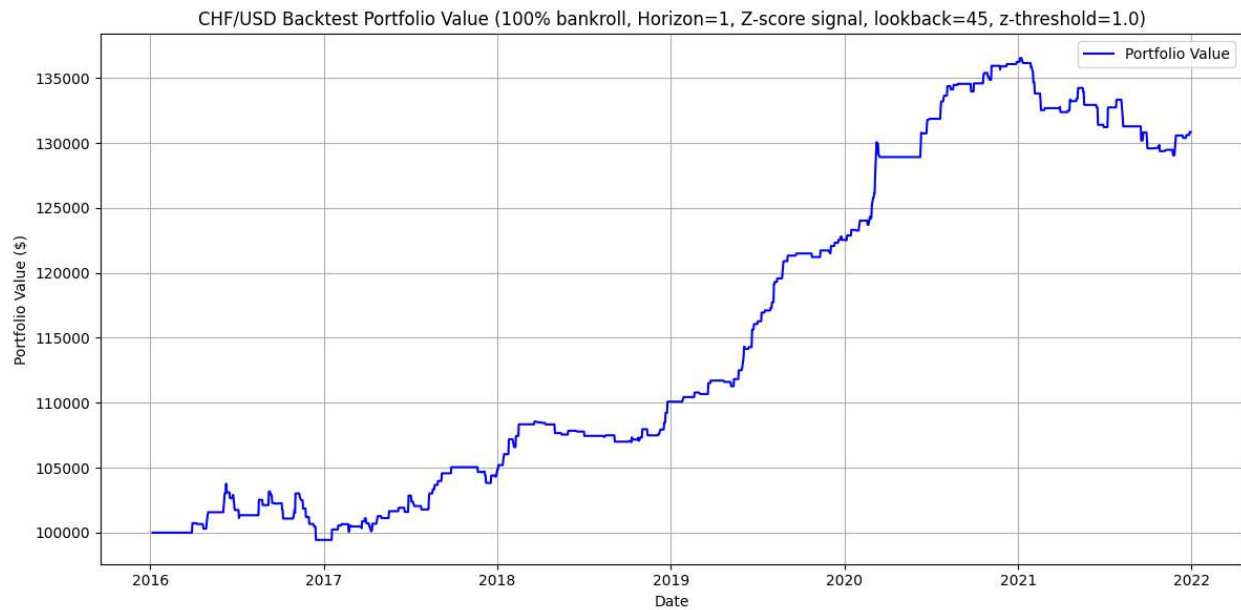
BEST 4% RFR:

Lookback: 40 days, Z-threshold: 0.80, Sharpe: 0.19, Trades: 282.0, Days in position: 282,

Cumulative Return: 32.52%

For robustness and considering number of days in trades, I did decide to use 45 day lookback at 1.0 z-score.

Training:



Cumulative return: 30.84%

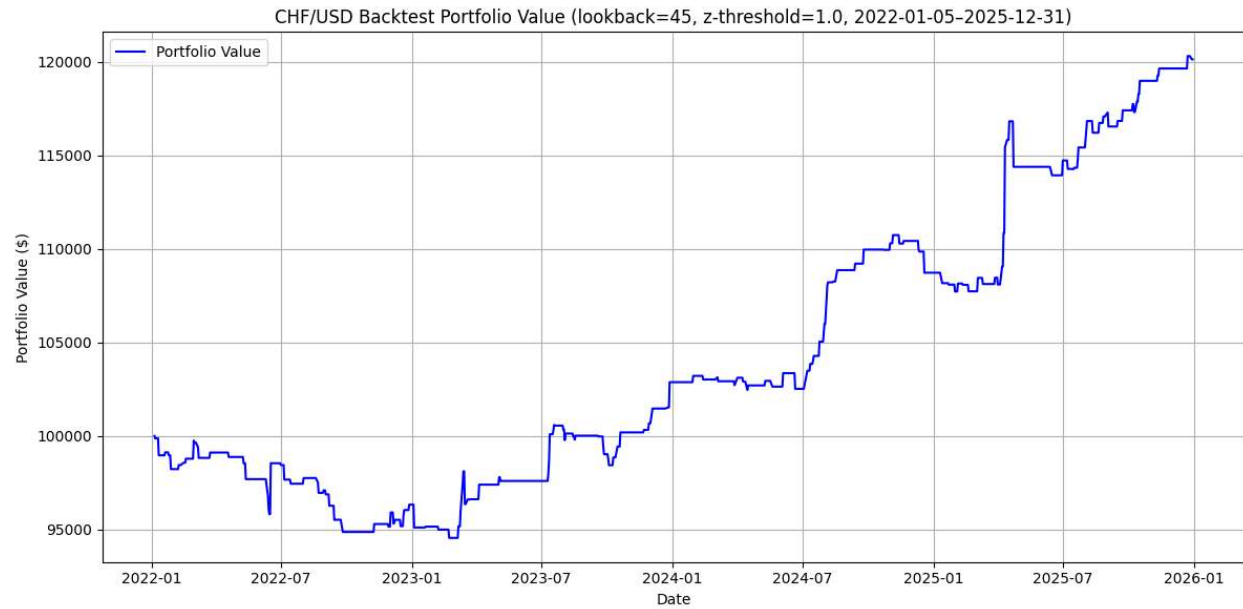
Annualized return: 4.43%

Annualized volatility: 3.07%

Sharpe ratio (0% RFR): 1.44

Sharpe ratio (4% RFR): 0.14

Testing:



Days in position: 159

Total trades: 128

Cumulative return: 20.12%

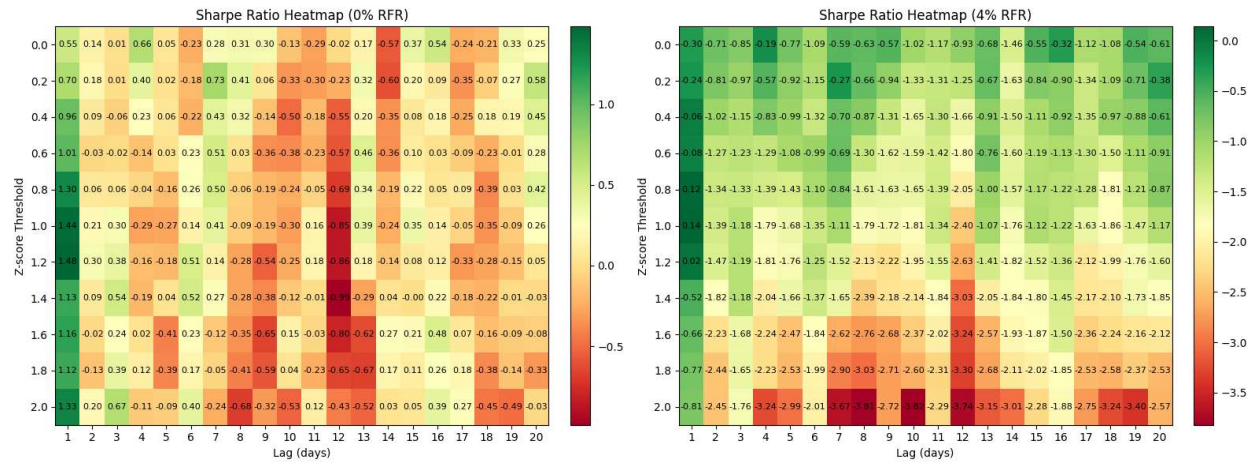
Annualized return: 4.73%

Annualized volatility: 4.44%

Sharpe ratio (0% RFR): 1.07

Sharpe ratio (4% RFR): 0.16

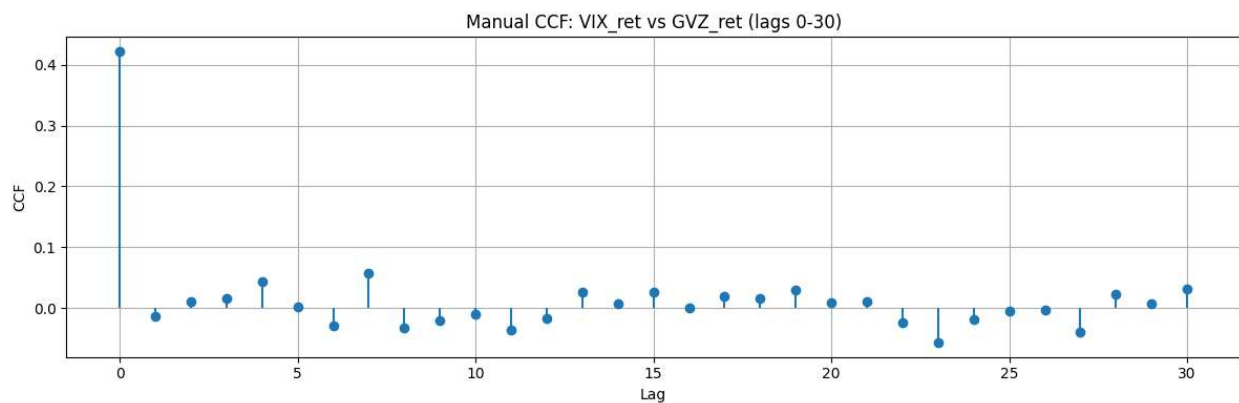
Just to make sure that 1 lag was the best, I ran a heatmap on lag and z-score with 45 day
lookback. If something other than lag 1 was consistently better, this would disprove my thesis as
it would be a value without statistical backing



Clearing 1-lag is the best, being robust across all z-scores and being relatively better than the rest.

VIX Filtered results

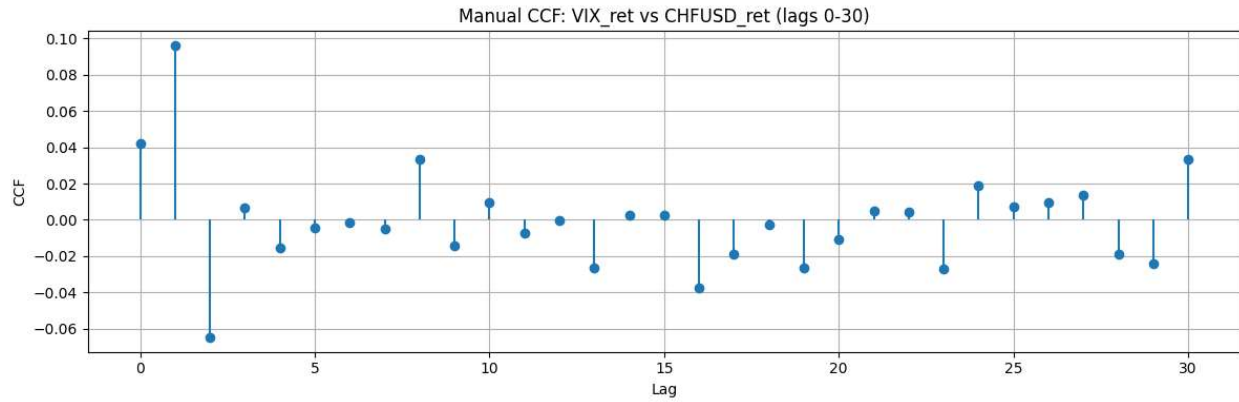
I decided to also add a filter on VIX to see how the strategy performs and seeing if this has effects. First I tested if VIX would even have any changes on results by running a CCF on VIX returns with both GVZ and CHF/USD returns



VIX_ret vs GVZ_ret

Lowest CCF: -0.0568 at lag 23

Highest CCF: 0.4217 at lag 0



VIX_ret vs CHFUSD_ret

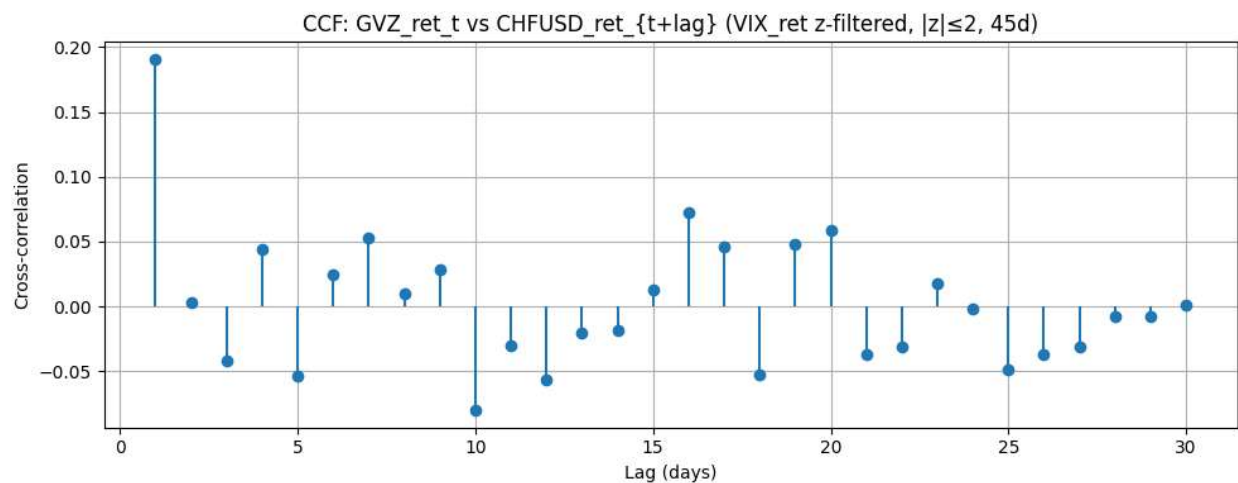
Lowest CCF: -0.0648 at lag 2

Highest CCF: 0.0961 at lag 1

These graphs shows that VIX effects GVZ with a lag of 0 and CHFUSD with a lag of 1. This mean VIX might be interfering with our signal.

For the following tests, I tested filtering out dates with VIX moves of 2 z-scores or less and saw if it yielded similar test results. (used a lookback of 45 days)

CCF test:

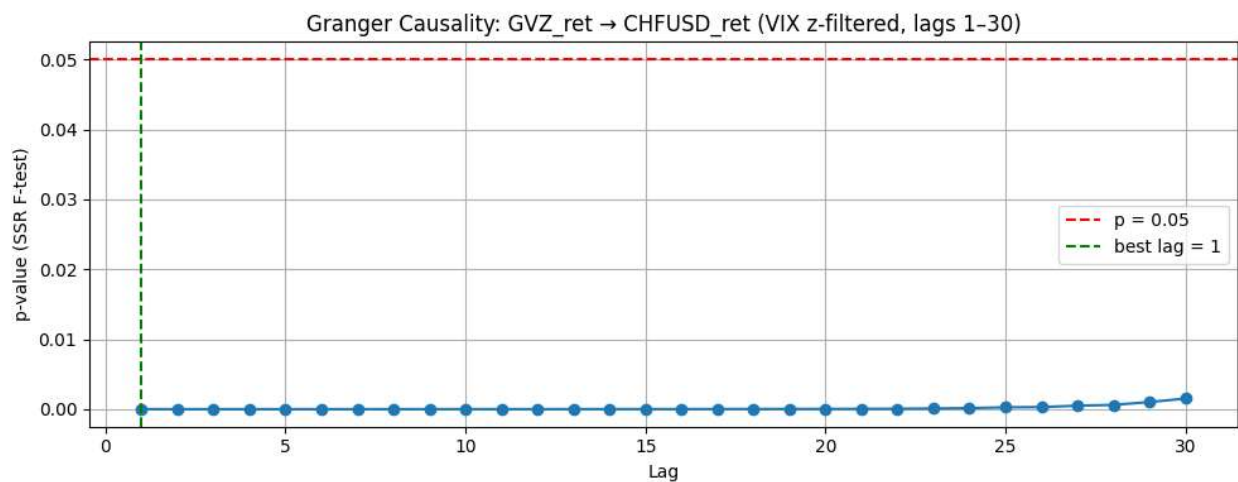


Highest correlation: 0.1904 at lag 1 days

Lowest correlation: -0.0799 at lag 10 days

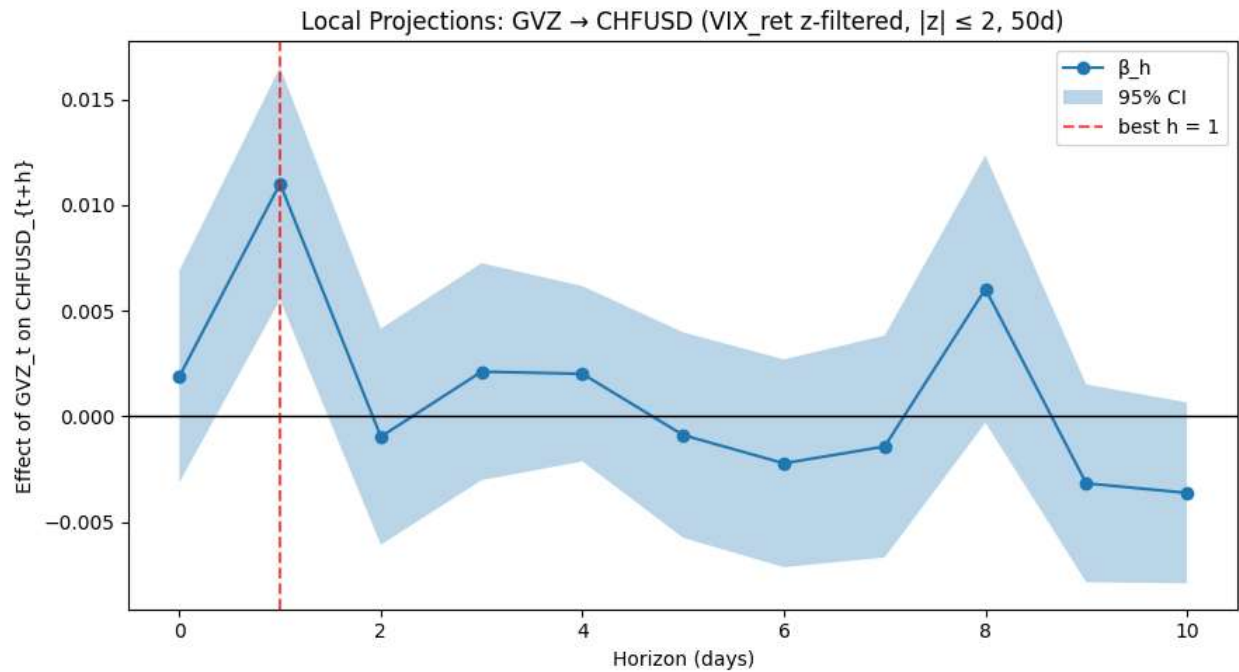
The CCF actually yields better results than before at a lag of 1.

Granger Causality test:



I believe the granger fails for a similar reason to before.

Local Projection:

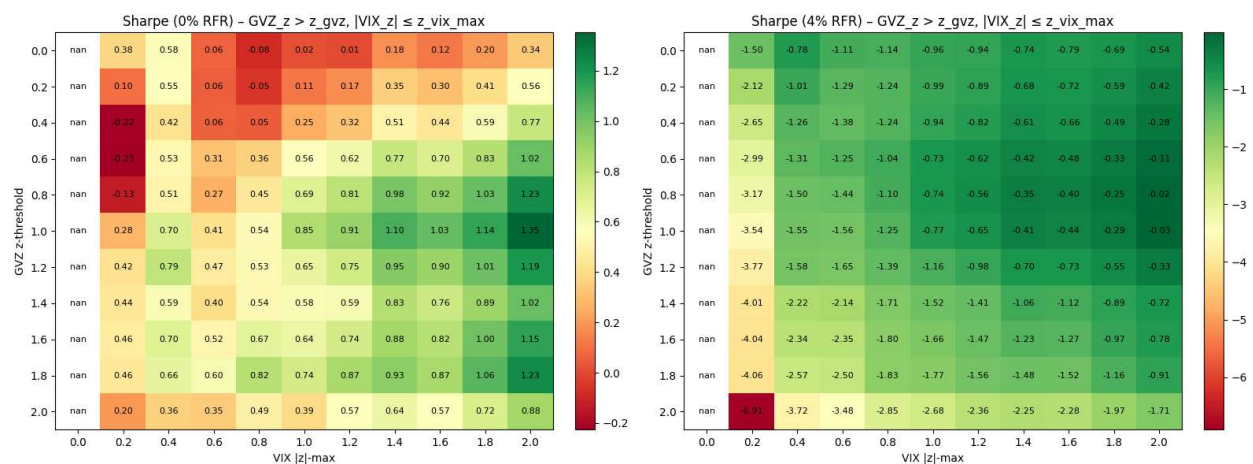


Horizon 1: Beta = 0.0113, stand error = 0.002847, t-score= 3.97, p-value = 7.183e-05

For a lag of 1 the Beta is significant.

I then optimized a sharpe for both z-score of VIX moves to filter out as well as z-scores of GVZ moves to enter. I used a lag of 1 and lookback of 45

Sharpe Heatmap:



BEST 0% RFR (GVZ z-threshold + VIX |z|-max):

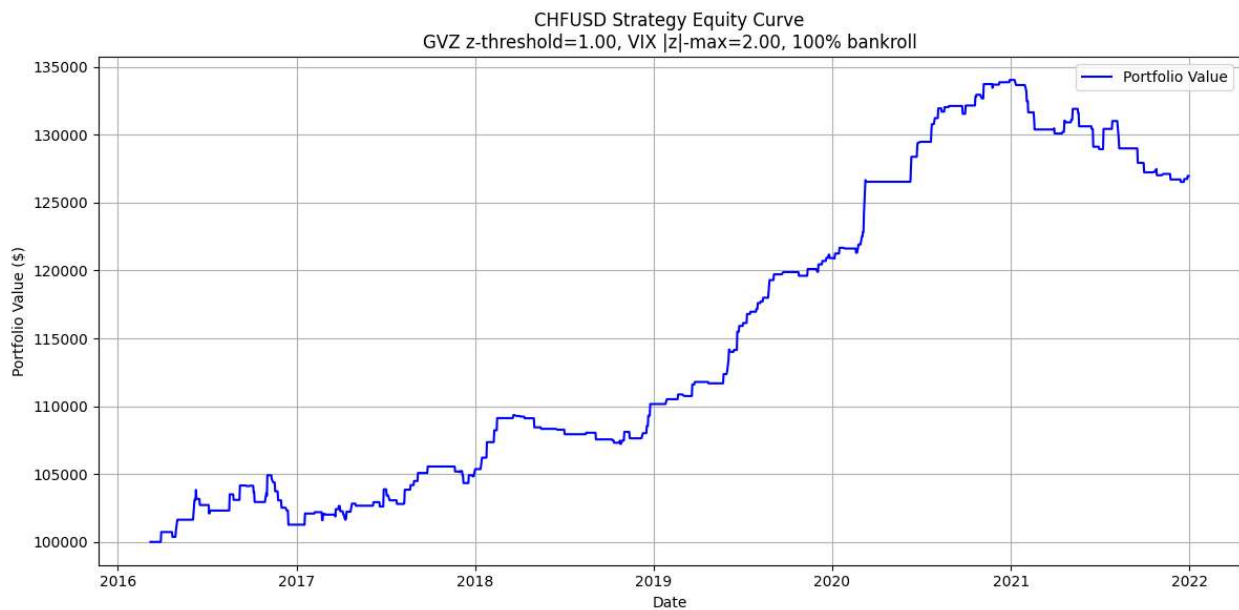
GVZ z-threshold: 1.00, VIX |z|-max: 2.00, Sharpe: 1.39, Trades: 189, Days in position: 189,

Cumulative Return: 26.97%

BEST 4% RFR (GVZ z-threshold + VIX |z|-max):

GVZ z-threshold: 1.00, VIX |z|-max: 2.00, Sharpe: 0.02, Trades: 189, Days in position: 189,

Cumulative Return: 26.97%



Total trades (days in position): 189

Days in position: 189

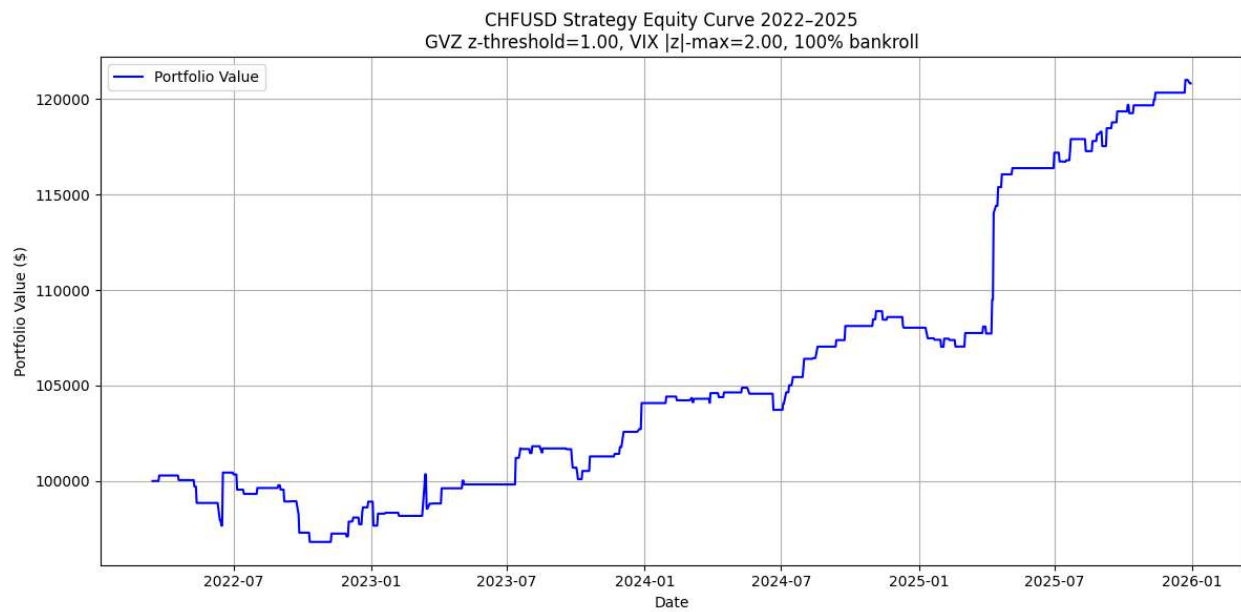
Cumulative return: 26.97%

Annualized return: 4.04%

Annualized volatility: 2.90%

Sharpe ratio vs 0% RFR: 1.39

Sharpe ratio vs 4% RFR: 0.02



Total trades (days in position): 126

Days in position: 126

Cumulative return: 20.82%

Annualized return: 4.94%

Annualized volatility: 3.90%

Sharpe ratio vs 0% RFR: 1.27

Sharpe ratio vs 4% RFR: 0.24